Pharmacode position may change as per Supplier's m/c requirement & additional small pharma code may appear on the front / back panel

MOXIWAR 400
P1537318

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P1537318

SUMMARY OF PRODUCT CHARACTERISTICS

MOXIWAR 400

Moxifloxacin Tablets 400 mg

R_x Only

1. NAME OF DRUG PRODUCT	:	Moxifloxacin Tablets 400 mg
(TRADE) NAME OF PRODUCT	:	MOXIWAR 400
STRENGTH	:	400 mg
PHARMACEUTICAL DOSAGE FORM	:	Film-coated Tablet.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION:

Each film-coated tablet contains Moxifloxacin Hydrochloride Ph.Eur. equivalent to Moxifloxacin 400 mg.

3. PHARMACEUTICAL FORM:

Dull red colored, modified capsule shaped film-coated tablets debossed with "E 18" on one side and plain on the other side.

ROUTE OF ADMINISTRATION: Oral

4. CLINICAL PARTICULARS:

4.1 Therapeutic indications:

Moxifloxacin tablets are indicated for the treatment of adults (> 18 years of age) with the following bacterial infections caused by susceptible strains:

- Acute sinusitis
- Acute exacerbation of chronic obstructive pulmonary disease including chronic bronchitis*
  *MOXIWAR 400 should be only used:
  When Pseudomonas is considered AND patient is allergic to antipseudomonal penicillins/cephalosporins;
  For resistant organisms with no other alternative antibiotics available.
- Community acquired pneumonia
- Mild to moderately severe inflammatory pelvic diseases (i.e. Infections of the upper female genital tract, including salpingitis and endometritis), without an associated tubo-ovarian or pelvic abscess. Moxifloxacin tablets 400 mg are not recommended for monotherapy of mild to moderately severe inflammatory pelvic diseases. Preferably, they should be administered in combination with another suited antibiotic (such as cephalosporin), due to the increasing resistance of Neisseria gonorrhoeae to moxifloxacin; that is, unless moxifloxacin-resistant Neisseria gonorrhoeae can be ruled out.
- Complicated skin and skin structure infections
- Complicated intra-abdominal infections including polymicrobial infections such as abscesses
- Consideration should be given to official guidance on the appropriate use of antibacterial agents.

4.2 Posology and method of administration:

Adults: The recommended dose Moxifloxacin is 400 mg once daily (1 film-coated tab) for the above mentioned indications and should not be exceeded.
Duration of Treatment: The duration of treatment should be determined by the severity of the indication or clinical response.
The following general recommendations for the treatment of upper and lower respiratory tract infections are made:
Bronchitis: acute exacerbation of chronic bronchitis, 5 days
Pneumonia: community acquired pneumonia, 10 days
Sinusitis: acute sinusitis, 7 days
Mild to moderately severe inflammatory pelvic diseases: 14 days
Complicated skin and skin structure infections:
Total treatment duration for sequential therapy (intravenous followed by oral therapy): 7 – 21 days
Complicated intraabdominal infections:
Total treatment duration for sequential therapy (intravenous followed by oral therapy): 5 - 14 days

The recommended duration of treatment for the indication being treated should not be exceeded.

Method of administration:
The tablet should be swallowed whole with sufficient liquid and may be taken independent of meals.

Geriatric patients:
No adjustment of dosage is required in the elderly.

Pediatric Patients:
The efficacy of Moxifloxacin in children and adolescents has not been established. No recommendation on posology can be made. The safety of Moxifloxacin in children below the age of 6 years has not been established.

Ethnic differences:
No adjustment of dosage is required in ethnic groups.

Patients with hepatic impairment:
No dosage adjustment is required in patients with impaired liver function

Patients with renal impairment:
No dose adjustment is required in patients with renal impairment (including creatinine clearance ≤ 30 ml/min/1.73m2) and in patients on chronic dialysis i.e. hemodialysis and continuous ambulatory peritoneal dialysis

4.3 Contraindications:

- Hypersensitivity to moxifloxacin or other quinolones, or any of the excipients of Moxifloxacin.
- Moxifloxacin must not be used in pregnant women
- breast-feeding is contraindicated during moxifloxacin therapy.
- Patients below 18 years of age.
- Patients with a history of tendon disease/disorder related to quinolone treatment

Contraindicated in patients with:

- Congenital or documented acquired QT prolongation
- Electrolyte disturbances, particularly in uncorrected hypokalaemia
- Clinically relevant bradycardia
- Clinically relevant heart failure with reduced left-ventricular ejection fraction
- Previous history of symptomatic arrhythmias

Moxifloxacin should not be used concurrently with other drugs that prolong the QT interval.

Moxifloxacin is also contraindicated in patients with impaired liver function (Child Pugh C) and in patients with transaminases increase > 5 fold ULN.

4.4 Special Warnings and precautions for Use:

The use of MOXIWAR 400 should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Adverse Effects/Undesirable Effects). Treatment of these patients with MOXIWAR 400 should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Prolonged, disabling and potentially irreversible serious adverse drug reactions
Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. MOXIWAR 400 should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Tendinitis and tendon rupture
Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids*. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with MOXIWAR 400 should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

*[For systemically administered levofloxacin-containing products, the listing of risk factors should additionally include: "in patients receiving daily doses of 1000 mg levofloxacin"].

The benefit of moxifloxacin treatment especially in infections with a low degree of severity should be balanced with the information contained in the warnings and precautions section

Prolongation of QTc interval and potentially QTc-prolongation-related clinical conditions
Moxifloxacin has been shown to prolong the QTc interval on the electrocardiogram in some patients. QTc prolongation with moxifloxacin was 6 msec ± 26 msec, 1.4% compared to baseline. As women tend to have a longer baseline QTc interval compared with men, they may be more sensitive to QTc-prolonging medications. Elderly patients may also be more susceptible to drug-associated effects on the QT interval.

Medication that can reduce potassium levels should be used with caution in patients receiving Moxifloxacin.

Moxifloxacin should be used with caution in patients with ongoing proarrhythmic conditions (especially women and elderly patients), such as acute myocardial ischaemia or QT prolongation as this may lead to an increased risk for ventricular arrhythmias (incl. torsade de pointes) and cardiac arrest. The magnitude of QT prolongation may increase with increasing concentrations of the drug. Therefore, the recommended dose should not be exceeded Medication that can reduce potassium levels should be used with caution in patients receiving moxifloxacin.

Moxifloxacin should be used with caution in patients with ongoing proarrhythmic conditions (especially women and elderly patients), such as acute myocardial ischaemia or QT prolongation as this may lead to an increased risk for ventricular arrhythmias (incl. torsade de pointes) and cardiac arrest. The magnitude of QT prolongation may increase with increasing concentrations of the drug. Therefore, the recommended dose should not be exceeded.

If signs of cardiac arrhythmia occur during treatment with moxifloxacin, treatment should be stopped and an ECG should be performed.

Hypersensitivity/allergic reactions
Hypersensitivity and allergic reactions have been reported for fluoroquinolones including moxifloxacin after first administration. Anaphylactic reactions can progress to a life-threatening shock, even after the first administration. In these cases moxifloxacin should be discontinued and suitable treatment (e.g. treatment for shock) initiated.

Severe liver disorders
Patients should be advised to contact their doctor prior to continuing treatment if signs and symptoms of fulminant hepatic disease develop such as rapidly developing asthenia associated with jaundice, dark urine, bleeding tendency or hepatic encephalopathy.

Liver function tests/investigations should be performed in cases where indications of liver dysfunction occur.

Serious bullous skin reactions
Bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis may occur with moxifloxacin. Patients should be advised to contact their doctor immediately prior to continuing treatment if skin and/or mucosal reactions occur.

Patients predisposed to seizures
Quinolones are known to trigger seizures. Use should be with caution in patients with CNS disorders or in the presence of other risk factors which may predispose to seizures or lower the seizure threshold. In case of seizures, treatment with moxifloxacin should be discontinued and appropriate measures instituted.

Peripheral neuropathy
Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with MOXIWAR 400 should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Adverse Effects/Undesirable Effects).

Psychiatric reactions
Psychiatric reactions may occur even after the first administration of fluoroquinolones, including moxifloxacin. In very rare cases depression or psychotic reactions can progressed to suicidal ideations/thoughts and self-injurious behavior, such as attempted or completed suicide (see section 'Undesirable effects'). In the event that the patient develops these reactions, moxifloxacin should be discontinued and appropriate measures instituted. Caution is recommended if moxifloxacin is to be used in psychotic patients or in patients with history of psychiatric disease.

Antibiotic-associated diarrhoea incl. colitis
Antibiotic-associated diarrhoea (AAD) and antibiotic-associated colitis (AAC), including pseudomembranous colitis and Clostridium difficile-associated diarrhoea, may ocur in association with the use of broad spectrum antibiotics including moxifloxacin and may range in severity from mild diarrhoea to fatal colitis. Therefore it is important to consider this diagnosis in patients who develop serious diarrhoea during or after the use of moxifloxacin. If AAD or AAC is suspected or confirmed, ongoing treatment with antibacterial agents, including moxifloxacin, should be discontinued and adequate therapeutic measures should be initiated immediately. Furthermore, appropriate infection control measures should be undertaken to reduce the risk of transmission. Drugs inhibiting peristalsis are contraindicated in patients who develop serious diarrhoea.

Exacerbation of myasthenia gravis
Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with flouroquinolones use in persons with myasthenia gravis. Avoid flouroquinolones in patients with known history of myasthenia gravis.

Tendon inflammation, tendon rupture
Tendon inflammation and rupture (especially Achilles tendon), sometimes bilateral, may occur with quinolone therapy including moxifloxacin, even within 48 hours of starting treatment. The risk of tendinitis and tendon rupture is increased in elderly patients and in those treated concurrently with corticosteroids. At the first sign of pain or inflammation, patients should discontinue treatment with moxifloxacin, rest the affected limb(s) and consult their doctor immediately in order to initiate appropriate treatment (e.g. immobilisation) for the affected tendon.

Patients with renal impairment
Elderly patients with renal disorders should use moxifloxacin with caution if they are unable to maintain adequate fluid intake, because dehydration may increase the risk of renal failure.

Vision disorders
If vision becomes impaired or any effects on the eyes are experienced, an eye specialist should be consulted immediately.

Dysglycemia
As with all fluoroquinolones, disturbances in blood glucose, including both hypoglycemia and hyperglycemia occurs with moxifloxacin. Dysglycemia occurrs predominantly in elderly diabetic patients receiving concomitant treatment with an oral hypoglycemic agent (e.g. sulfonylurea) or with insulin. In diabetic patients, careful monitoring of blood glucose is recommended.

Prevention of photosensitivity reactions
Quinolones have been shown to cause photosensitivity reactions in patients. Moxifloxacin has a lower risk to induce photosensitivity. Patients should be advised to avoid exposure to either UV irradiation or extensive and/or strong sunlight during treatment with moxifloxacin.

Patients with glucose-6-phosphate dehydrogenase deficiency
Patients with a family history of, or actual glucose-6-phosphate dehydrogenase deficiency are prone to haemolytic reactions when treated with quinolones. Therefore, moxifloxacin should be used with caution in these patients.

Patients with galactose intolerance, Lapp lactase deficiency or glucose-galactose malabsorption
Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Patients with pelvic inflammatory disease
For patients with complicated pelvic inflammatory disease (e.g. associated with a tubo-ovarian or pelvic abscess), for whom an intravenous treatment is considered necessary, treatment with Moxifloxacin tablets 400 mg is not recommended.

Pelvic inflammatory disease may be caused by fluoroquinolone-resistant Neisseria gonorrhoeae. Therefore in such cases empirical moxifloxacin should be co-administered with another appropriate antibiotic (e.g. a cephalosporin) unless moxifloxacin-resistant Neisseria gonorrhoeae can be excluded.

Interference with biological tests
Moxifloxacin therapy may interfere with the Mycobacterium spp. culture test by suppression of mycobacterial growth causing false negative results in samples taken from patients currently receiving moxifloxacin.

Patients with MRSA infections
Moxifloxacin is not recommended for the treatment of MRSA infections. In case of a suspected or confirmed infection due to MRSA, treatment with an appropriate antibacterial agent should be started.

Paediatric population
Due to adverse effects on the cartilage in juvenile animals the use of moxifloxacin in children and adolescents < 18 years is contraindicated.

Aortic aneurysm and dissection
Epidemiologic studies report and increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis)

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

4.5 Interaction with other medicinal products and other forms of interaction:

Interactions with medicinal products
An additive effect on QT interval prolongation of moxifloxacin and other medicinal products that may prolong the QT interval cannot be excluded. This might lead to an increased risk of ventricular arrhythmias, including torsade de pointes.

Therefore, co-administration of moxifloxacin with any of the following medicinal products is contraindicated:

- anti-arrhythmics class IA (e.g. quinidine, hydroquinidine, disopyramide)
- anti-arrhythmics class III (e.g. amiodarone, sotalol, dofetilide, ibutilide)
- antipsychotics (e.g. phenothiazines, pimozide, sertindole, haloperidol, sultopride)
- tricyclic antidepressive agents
- certain antimicrobial agents (saquinavir, sparfloxacin, erythromycin IV, pentamidine, antimalarials particularly halofantrine)
- certain antihistaminics (terfenadine, astemizole, mizolastine)
- others (cisapride, vincamine IV, bepridil, diphemanil).

Moxifloxacin should be used with caution in patients who are taking medication that can reduce potassium levels (e.g. loop and thiazide-type diuretics, laxatives and enemas [high doses], corticosteroids, amphotericin B) or medication that is associated with significant bradycardia.

An interval of about 6 hours should be left between administration of agents containing bivalent or trivalent cations (e.g. antacids containing magnesium or aluminium, didanosine tablets, sucralfate and agents containing iron or zinc) and administration of moxifloxacin.

Concomitant administration of charcoal with an oral dose of 400 mg Moxifloxacin is not recommended.

No precaution is required for use with digoxin.

Concomitant administration of oral moxifloxacin with glibenclamide resulted in a decrease in the peak plasma concentrations of glibenclamide. The combination of glibenclamide and moxifloxacin could theoretically result in a mild and transient hyperglycaemia. However, the observed pharmacokinetic changes for glibenclamide did not result in changes of the pharmacodynamic parameters (blood glucose, insulin). Therefore no clinically relevant interaction was observed between moxifloxacin and glibenclamide.

Changes in INR
An increase in oral anticoagulant activity have been reported in patients receiving antibacterial agents, especially fluoroquinolones, macrolides, tetracyclines, cotrimoxazole and some cephalosporins. The infectious and inflammatory conditions, age and general status of the patient appear to be risk factors. Under these circumstances, it is difficult to evaluate whether the infection or the treatment caused the INR (international normalised ratio) disorder. A precautionary measure would be to more frequently monitor the INR. If necessary, the oral anticoagulant dosage should be adjusted as appropriate.

No interactions have been found following concomitant administration of moxifloxacin with: ranitidine, probenecid, oral contraceptives, calcium supplements, morphine administered parenterally, theophylline, cyclosporine or itraconazole.

Interaction with food
Moxifloxacin has no relevant interaction with food including dairy products.

4.6 Pregnancy and lactation

Pregnancy
The safe use of moxifloxacin in human pregnancy has not been established. Reversible joint injuries are described in children receiving some quinolones, however, this effect has not been reported as occurring on exposed foetuses. Animal studies have shown reproductive toxicity. The potential risk for human is unknown. Consequently, the use of moxifloxacin during pregnancy is contraindicated.

Lactation
As with other quinolones, moxifloxacin has been shown to cause lesions in the cartilage of the weight bearing joints of immature animals. Preclinical evidence indicates that small amounts of moxifloxacin may be secreted in human milk. There is no data available in lactating or nursing women. Therefore, the use of moxifloxacin in nursing mothers is contraindicated.

4.7 Effects on Ability to Drive and Use Machine

Fluoroquinolones including moxifloxacin may result in an impairment of the patient's ability to drive or operate machinery due to CNS reactions and vision disorders (see section Undesirable Effects).

4.8 Undesirable effects:

Adverse reactions with moxifloxacin 400 mg (oral and sequential therapy) sorted by frequencies are listed below:

System Organ Class (MedDRA)	Common	Un common	Rare	Very Rare
Infections and infestations	Super infections due to resistant bacteria or fungi e.g. oral and vaginal candidiasis			
Blood and lymphatic system disorders		Anaemia Leucopenia(s) Neutropenia Thrombocytopenia Thrombocythemia Blood eosinophilia Prothrombin time prolonged/INR increased		Prothrombin level increased/INR decreased Agranulocytosis
Immune system disorders		Allergic reaction	Anaphylaxis incl. very rarely life-threatening shock Allergic oedema / angio oedema (incl. laryngeal oedema, potentially life threatening.	
Metabolism and nutrition disorders		Hyperlipidemia	Hyperglycemia Hyperuricemia	Hypoglycemia
Psychiatric disorders		Anxiety reactions Psychomotor hyperactivity/ agitation	Depression (in very rare cases potentially culminating in self-injurious behaviour, such as suicidal ideation/ thoughts or suicide attempts), Hallucination	Psychotic reactions (potentially culminating in self-injurious behaviour, such as suicidal ideation/ thoughts or suicide attempts)
Nervous system disorders	Headache Dizziness	Par- and Dysaesthesia Taste disorders (incl. ageusia in very rare cases) Confusion and disorientation Sleep disorders (predominantly insomnia) Tremor Vertigo Somnolence	Hypoaesthesia Smell disorders (incl. anosmia) Abnormal dreams Disturbed coordination (incl. gait disturbances, esp. due to dizziness or vertigo) Seizures incl. grand mal convulsions Disturbed attention Speech disorders Amnesia Peripheral neuropathy and polyneuropathy	Hyperaesthesia
Eye disorders		Visual disturbances incl. diplopia and blurred vision (especially in the course of CNS reactions)		Transient loss of vision (especially in the course of CNS reactions)
Ear and labyrinth disorders			Tinnitus Hearing impairment incl. deafness (usually reversible)	
Cardiac disorders	QT prolongation in patients with hypokalaemia	QT prolongation (see Warning & Precaution) Palpitations Tachycardia Atrial fibrillation Angina pectoris	Ventricular tachyarrhythmias Syncope (i.e., acute and short lasting loss of consciousness)	Unspecified Arrhythmias Torsade de Pointes Cardiac arrest
Vascular disorders		Vasodilatation	Hypertension Hypotension	Vasculitis
Respiratory, thoracic and mediastinal disorders		Dyspnea (including asthmatic conditions)		
Gastrointestinal disorders	Nausea Vomiting Gastrointestinal and abdominal pains Diarrhoea	Decreased appetite and food intake Constipation Dyspepsia Flatulence Gastritis Increased amylase	Dysphagia Stomatitis Antibiotic associated colitis (incl. pseudomembranous colitis, in very rare cases associated with life threatening complications,	
Hepatobiliary disorders	Increase in transaminases	Hepatic impairment (incl. LDH increase) Increased bilirubin Increased gammaglutamyl-Transferase Increase in blood alkaline phosphatase	Jaundice Hepatitis (predominantly cholestatic)	Fulminant hepatitis potentially leading to life-threatening liver failure (incl. fatal cases)
Skin and subcutaneous tissue disorders		Pruritus Rash Urticaria Dry skin		Bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis (potentially life-threatening)
Musculoskeletal and connective tissue disorders		Arthralgia Myalgia	Tendonitis (see Warning & Precaution) Muscle cramp Muscle twitching Muscle weakness	Tendon rupture (see Warning & Precaution) Arthritis Muscle rigidity Exacerbation of symptoms of myasthenia gravis
Renal and urinary disorders		Dehydration	Renal impairment (incl. increase in BUN and creatinine) Renal failure	
General disorders and administration site conditions		Feeling unwell (predominantly asthenia or fatigue) Painful conditions (incl. pain in back, chest, pelvic and extremities) Sweating	Oedema	

ADVERSE REACTIONS/SIDE EFFECTS

<u>Exacerbation of myasthenia gravis</u>
Post Marketing Experience

ADVERSE EFFECTS/UNDESIRABLE EFFECTS:

Musculoskeletal and connective tissue disorders*
Nervous system disorders*
General disorders and administrative site conditions*
Psychiatric disorders*
Eye disorders*
Ear and labyrinth disorders*

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

4.9 Symptoms and Treatment of Overdose

No specific countermeasures after accidental overdose are recommended. In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be undertaken, because of the possibility of QT interval prolongation. Concomitant administration of charcoal with a dose of 400 mg oral moxifloxacin will reduce systemic availability of the drug by more than 80%. The use of charcoal early during absorption may be useful to prevent excessive increase in the systemic exposure to moxifloxacin in cases of oral overdose.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamics

Pharmacotherapeutic group: Quinolone antibacterials, fluoroquinolones, ATC code: J01MA14

Mechanism of action

Moxifloxacin has in vitro activity against a wide range of Gram-positive and Gram-negative pathogens.

The bactericidal action of moxifloxacin results from the inhibition of both type II topoisomerases (DNA gyrase and topoisomerase IV) required for bacterial DNA replication, transcription and repair.

It appears that the C8- methoxy moiety contributes to enhanced activity and lower selection of resistant mutants of Grampositive bacteria compared to the C8-H moiety. The presence of the bulky bicycloamine substituent at the C-7 position prevents active efflux, associated with thenorA or pmrA genes seen in certain Grampositive bacteria. Moxifloxacin exhibits a concentration dependent killing rate. Minimum bactericidal concentrations (MBC) were found to be in the range of the minimum inhibitory concentrations (MIC).

Effect on the intestinal flora in humans

The following changes in the intestinal flora were seen following oral administration of moxifloxacin:
Escherichia coli, Bacillus spp., Enterococcus spp., and Klebsiella spp. Reduced, as were the anaerobes Bacteroides vulgatus, Bifidobacterium spp., Eubacterium spp., and Peptostreptococcus spp.. For Bacteroides fragilis there was an increase. These changes returned to normal within two weeks.

Mechanism of resistance

Resistance mechanisms that inactivate penicillins, cephalosporins, aminoglycosides, macrolides and tetracyclines do not interfere with the antibacterial activity of moxifloxacin. Other resistance mechanisms such as permeation barriers (common in Pseudomonas aeruginosa) and efflux mechanisms may also effect susceptibility to moxifloxacin. In vitro resistance to moxifloxacin is acquired through a stepwise process by target site mutations in both type II topoisomerases, DNA gyrase and topoisomerase IV. Moxifloxacin is a poor substrate for active efflux mechanisms in Gram-positive organisms. Cross resistance is observed with other fluoroquinolones. However, as moxifloxacin inhibits both topoisomerase II and IV with similar activity in some Grampositive bacteria, such bacteria may be resistant to other quinolones, but susceptible to moxifloxacin.

5.2 Pharmacokinetic properties:

Absorption and Bioavailability

Following oral administration, moxifloxacin is absorbed rapidly and almost completely. The absolute bioavailability amounts to approximately 91%.

Distribution

Moxifloxacin is distributed very rapidly to extravascular spaces rapidly. The steady-state volume of distribution (Vss) is approximately 2 l/kg. Moxifloxacin is mainly bound to serum albumin.

Biotransformation

Moxifloxacin undergoes phase II biotransformation and is excreted via renal and biliary/faecal pathways as unchanged drug as well as in form of a sulfo-compound (M1) and a glucuronide (M2). M1 and M2 are the only metabolites relevant in humans, both are microbiologically inactive.

Elimination

Moxifloxacin is eliminated from plasma with a mean terminal half-life (t½) of approximately 12 hrs. The mean apparent total body clearance following a 400 mg dose ranges from 179 to 246 ml/min. Renal clearance amounted to about 24 - 53 ml/min suggesting partial tubular reabsorption of the drug from the kidneys.

Elderly and patients with low body weight

Higher plasma concentrations are observed in low body weight (such as women) and in elderly

Renal impairment

The pharmacokinetic properties of moxifloxacin are not significantly different in patients with renal impairment (including creatinine clearance >20 ml/min/1.73m2). As renal function decreases, concentrations of the M2 metabolite (glucuronide) increase by up to a factor of 2.5 (with a creatinine clearance of < 30 ml/min/1.73m2).

Hepatic impairment

There is insufficient experience in the clinical use of moxifloxacin in patients with impaired liver function.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Microcrystalline cellulose (PH 101 and PH 102), Sodium starch glycolate, Povidone, Magnesium stearate, Opadry pink 03B54025 & Purified water.

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

60 months.

6.4 Special precautions for storage

Store at below 30°C.

6.5 Nature and contents of container

Blister pack: 2 Blisters of each 5 tablets (2x5's).

7. MANUFACTURED BY:

Aurobindo Pharma Limited, Unit-VII (SEZ)
Special Economic Zone Tsiic,
Plot No S1 Sy Nos 411/P 425/P 434/P
435/P and 458/P Green Industrial Park,
Jadcherla Mandal Mahabubnagar District,
Polepally Village, 509302, India.

8. Product Registration Holder

Healol Pharmaceuticals Sdn. Bhd.
74-3, Jalan Wangsa Delima 6,
KLSC Wangsa Maju, 53300,
Kuala Lumpur, Malaysia.

9. DATE OF PREPARATION OF THIS LEAFLET

June 2024.

P1537318

AUROBINDO Packaging Development	Product Name: MOXIWAR 400	Component: Leaflet	Item Code: P1537318	Date & Time: 06.11.2024 & 05:39 PM
	Customer / Country: Malaysia_Unit-NA	Version No.: 03	Reason of Issue: New	No. of Colours : **1** Black
Team Leader: Kirankumar K Initiator: Krishna P	Dimensions: A/s: 280 x 420 mm B/s: 35 x 50 mm	Font Type: Arial	Font Size: 6 pt	
Design Agency: vakils PREMEDIA	Pharmacode: 37318			

Date - Designer - Run:
1. 25-JUN-2024 (Rupesh) - FR
2. 09-JUL-2024 (Ram) - RR1
3. 18-OCT-2024 (Adi) - RR2
4. 06-NOV-2024 (Dipali) - RR3
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Reviewed / Approved by

Sign / Date

Additional Information: Supercede Item Code: NIL